OrangePi Zero3 LTE router: simei user guide

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This guide explains how to operate the OrangePi Zero3 router (OpenWrt 25.12.1) with the Quectel EG25-G modem and its embedded eUICC: how to reach the router, how to manage eSIM profiles, and how to keep one IMEI per SIM profile with the **simei** tool.

Changing an IMEI is illegal in some countries. simei writes whatever IMEI you assign to a SIM profile into your own modem; the responsibility for using it is yours.

SECTION

1. Reaching the router

The board joins the home LAN as a DHCP client with hostname `orangepi-lte`. Its current address is shown by the home router's DHCP table; from the Mac it is kept in `tools/router.env`. Over Tailscale the board is always `orangepi-lte` (100.111.153.32).

REQUIREMENT

1.1. Web interface (LuCI)

UID: ACCESS-1

STATEMENT:
Open `http://<router-ip>/` in a browser. Log in as `root` with an empty password until a password is set (LuCI > System > Administration). Use plain `http`: the web TUI is embedded over http and browsers refuse to embed it inside an https page.

REQUIREMENT

1.2. ssh

UID: ACCESS-2

STATEMENT:
`ssh root@<router-ip>` works without a password from the Mac (key `~/.ssh/id_ed25519`) and from the phone with the key `~/.ssh/orangepi_phone_ed25519` (import the private key into the phone's ssh client). Over the tailnet: `ssh root@orangepi-lte` (Tailscale SSH is enabled, no key needed from tailnet devices).

REQUIREMENT

1.3. Web terminals

UID: ACCESS-3

STATEMENT:
`http://<router-ip>:7681/` opens simei directly in the browser (also reachable as LuCI > Modem > SIM / IMEI (TUI)). `http://<router-ip>:7682/` opens a root shell in the browser. Both are reachable from the LAN and the tailnet only.

REQUIREMENT

1.4. Raw AT commands from another computer

UID: ACCESS-4

STATEMENT:
The modem's AT port is exposed on TCP port 5000 by the `atbridge` service: `socat - TCP:<router-ip>:5000` then type `AT+GSN` (Enter). One client at a time.

REQUIREMENT

1.5. Serial console

UID: ACCESS-5

STATEMENT:
With the USB serial adapter connected to the Mac: `tools/serial.sh console` (exit with Ctrl-]), or `tools/serial.sh run "ip addr"` for one command. picocom must be closed first.

SECTION

2. simei: concepts

simei keeps a **profile** for every SIM you own. A profile is identified by the SIM's ICCID and holds a name, the IMEI to use with that SIM, and optionally an APN. Profiles live in `/etc/config/simei`.

The modem carries an eUICC, so "inserting another SIM" means **enabling another eSIM profile**. simei talks to the eUICC through `lpac` and to the modem through the AT bridge.

Two operations do the real work:

- `apply` looks at the SIM the modem currently sees, finds its profile, and if the profile's IMEI differs from the modem's IMEI it writes the IMEI (`AT+EGMR=1,7`), restarts the modem (`AT+CFUN=1,1`) and verifies the result.
- `switch` enables an eSIM profile, restarts the modem and then runs `apply` for it.

Reassigning an IMEI between two SIMs is therefore: `move` (or `swap`) the IMEI between the two profiles, then `switch` to (or `apply` on) the SIM that should carry it now.

SECTION

3. Using the TUI

REQUIREMENT

3.1. Starting the TUI

UID: TUI-1

STATEMENT:
Run `simei` over ssh, on the serial console, or open `http://<router-ip>:7681/`. The main menu offers: Status, Live dashboard, SIM profiles, Register the inserted SIM, Reassign IMEI, Apply, eSIM profiles, AT command, Restart, Exit. Menus are numbered; type the number or the name and press Enter (on a dumb console), or use the arrow keys (whiptail).

REQUIREMENT

3.2. Status

UID: TUI-2

STATEMENT:
Shows the modem IMEI (flagged when blank or invalid), SIM ICCID and IMSI, SIM state, signal, operator and access technology, registration state, and the matched profile with whether its IMEI matches the modem.

REQUIREMENT

3.3. Live dashboard

UID: TUI-3

STATEMENT:
Refreshes every second: download and upload rate of `wwan0` and session totals (read from the kernel counters, no modem round trips), LTE signal (RSSI, RSRP, RSRQ, SNR, from QMI every 3 s), IMEI, ICCID, operator. Leave with Enter (or any key where the terminal supports raw mode).

REQUIREMENT

3.4. eSIM profiles menu

UID: TUI-4

STATEMENT:
Lists every profile on the eUICC with its state, name/nickname and the IMEI assigned to it. Selecting a profile offers: **Switch** (enable it, restart the modem, write its IMEI), Enable, Disable, **Assign / change the IMEI for this SIM**, Set nickname, Delete. The menu also offers **Download a new profile** (activation code `LPA:1$<smdp>$<matching-id>` and, if the operator gave one, a confirmation code), eUICC info (EID, free memory) and **Diagnose the eSIM path**, which runs the same checks as `simei esim doctor`.

REQUIREMENT

3.5. Reassign IMEI

UID: TUI-5

STATEMENT:
Choose Move (source loses the IMEI, target receives it) or Swap, then the two profiles. If the SIM currently active is affected, simei offers to write the IMEI to the modem immediately.

REQUIREMENT

3.6. Apply

UID: TUI-6

STATEMENT:
Shows the planned change (`IMEI change: <old> -> <new>`) and asks for confirmation before writing. The modem restarts; the screen reports `verified: modem IMEI is now ...` when done.

SECTION

4. Using the command line

REQUIREMENT

4.1. Command reference

UID: CLI-1

STATEMENT:

```
simei status [--raw]           modem + SIM status, matched profile
simei live [--once]            live bandwidth / signal dashboard
simei list                     SIM profiles
simei register [name]          profile for the SIM the modem sees now
simei add <name> <iccid> [imei] [apn]
simei set <profile> name|imei|apn|iccid <value>
simei delete <profile>
simei move <from> <to>         hand the IMEI to another profile
simei swap <a> <b>             exchange the IMEIs
simei apply [--yes] [--dry-run]
simei switch <iccid|profile> [--yes]
simei esim [list] | info | enable <iccid> | disable <iccid>
simei esim delete <iccid> [--yes] | nickname <iccid> <name>
simei esim download 'LPA:1$smdp$id' [confirmation-code]
simei esim doctor [--debug]    check the eSIM path step by step
simei at "AT+..."              raw AT command
simei luhn <14|15 digits>      IMEI check digit
simei restart modem|wwan|bridge
simei usbmode [--fix]          USB mode (usbnet 0 = QMI); --fix switches an
```

REQUIREMENT

4.2. eSIM backends and diagnostics

UID: CLI-6

STATEMENT:
simei reaches the eUICC through `lpac`, over one of three backends set in `/etc/config/simei` as `esim_backend`:

- `at` (the default) uses the modem's second AT port, `esim_tty`, normally `/dev/ttyUSB3`;
- `uqmi` uses the QMI device, `qmi_dev`, normally `/dev/cdc-wdm0`;
- `auto` tries uqmi first and falls back to `at`.

Before the uqmi path is used, simei checks that `uqmi` is installed, that the QMI device is a character device whose path is a plain path, and that the modem still gives out a UIM client id. A failing check is reported by name instead of arriving as lpac's bare `euicc_init`. Only one eSIM operation runs at a time, and `euicc_init` / no-answer failures are retried `esim_retries` times (1 by default); an answer from the eUICC itself, such as a refused delete, is reported as it is.

`simei esim doctor` walks the whole path and prints one line per step: lpac binary, `jq`, the configured backend, the AT port or the QMI device, the UIM client id, the card state, and finally a real `chip info`. It exits 0 only when the eUICC answered. `--debug` adds lpac's raw output, including the uqmi driver's shell traffic.

```
simei esim doctor
uci set simei.global.esim_backend=auto; uci commit simei   # uqmi first, AT
```

REQUIREMENT

4.3. First SIM (the DENT eSIM already on the eUICC)

UID: CLI-2

STATEMENT:

```
simei register DENT                # creates sim1 for ICCID 89480100100871
simei set sim1 imei 35xxxxxxxxxxxx  # 14 digits: the check digit is append
simei apply                         # writes it, restarts the modem, verif
```

The modem shipped with IMEI 000000000000000 (blank); simei refuses to store or back up that value, so an IMEI has to be assigned once before the first `apply`.

REQUIREMENT

4.4. Adding a second eSIM and moving between them

UID: CLI-3

STATEMENT:

```
simei esim download 'LPA:1$smdp.example.com$ABCD-1234'   # from the operato
simei esim list                                          # note the new ICC
simei add "Work" <iccid> 35yyyyyyyyyyyy                  # its own IMEI
simei switch <iccid>                                     # enable + restart
simei switch sim1                                        # back to DENT
```

To give the second SIM the IMEI of the first: `simei move sim1 sim2` then `simei switch sim2`.

REQUIREMENT

4.5. Exit codes and logs

UID: CLI-4

STATEMENT:
`apply` / `switch`: 0 done or nothing to do; 1 cancelled or usage error; 2 the IMEI fails the 15-digit Luhn check; 3 no SIM, no profile, or no IMEI assigned; 4 the modem refused the write or did not come back; 5 the AT bridge or lpac is unreachable. Every write is logged to the system log (`logread | grep simei`). The first valid IMEI seen is saved to `/etc/simei.factory-imei`.

SECTION

5. Automatic mode

REQUIREMENT

5.1. auto_apply

UID: AUTO-1

STATEMENT:
About 20 s after the modem's AT port appears (boot, modem restart) the hotplug script `/etc/hotplug.d/tty/90-simei` runs `simei usbmode`: if the EG25-G is not in QMI USB mode (`AT+QCFG="usbnet"` other than 0, seen after a power loss, which leaves no `wwan0`) it runs `simei usbmode --fix`, which sets QMI mode and restarts the modem once; the port re-appears and the script runs again. `uci set simei.global.auto_apply=1; uci commit simei` additionally makes it run `simei apply --yes` afterwards, so the profile of whatever SIM is active gets its IMEI written without asking. Default for auto_apply is off; the USB mode guard is always on.

SECTION

6. Other modem tools on the router

- LuCI > Modem > **3ginfo-lite**: signal, band, cell, operator, connection details.
- LuCI > Modem > **SMS tool**: read and send SMS, USSD, send AT commands from the browser.
- LuCI > Network > Interfaces > **wwan**: the LTE connection (QMI, APN `internet`); change the APN there if the operator needs another one.
- `uqmi -d /dev/cdc-wdm0 --get-signal-info` and `--get-serving-system` from a shell.
- LuCI > Modem > **Preferred LTE/5G Bands**: band locking through `modemband` (both packages come from the 4IceG apk repository, installed by `tools/05-modemband.sh`); from a shell `modemband.sh getbands` and `modemband.sh setbands "3 7 20"`. The raw equivalent is `simei at 'AT+QCFG="band",0,0x80000,0'` (see the Quectel EG25-G AT manual). modemband and the SMS tool use `/dev/ttyUSB2`, the port behind the AT bridge: one client at a time.

SECTION

7. Maintenance from the Mac

REQUIREMENT

7.1. Updating simei on the router

UID: MAINT-1

STATEMENT:

```
sh simei/tests/run.sh          # local tests (socat, jq required)
tools/rt push simei /          # copy the tree to the router
tools/rt 'sh /usr/lib/simei/tests/run.sh'   # tests on the router
```

Everything is versioned in the private repository `github.com/jftochka/owrt-orangepi-lte`.

REQUIREMENT

7.2. Re-creating the router from scratch

UID: MAINT-2

STATEMENT:

Flash the official OpenWrt 25.12.1 image for xunlong_orangepi-zero3, connect serial and Ethernet, then run in order: `tools/01-bootstrap-net.sh`, `tools/rt - < tools/02-install.sh`, `tools/rt - < tools/05-modemband.sh` (4IceG apk repository: `modemband` + LuCI band page), `tools/rt push simei /`, `tools/rt - < tools/03-configure.sh`, `tools/rt - < tools/04-tailscale.sh` (open the printed login URL once). If the modem is in ECM mode (`AT+QCFG="usbnet"` returns 1) switch it to QMI with `AT+QCFG="usbnet",0` and `AT+CFUN=1,1`.

SECTION

8. Troubleshooting

- **simei: cannot reach the AT bridge**: `/etc/init.d/atbridge restart`; check `ls /dev/ttyUSB*`.
- **Registration: searching / No Service**: `simei at 'AT+QENG="servingcell"'` shows whether a cell is seen; a blank IMEI can be rejected by the network - assign one and `simei apply`.
- **lpac: euicc_init**: run `simei esim doctor` first - it names the failing step. Then check that the modem sees the eUICC at all: `simei at 'AT+QSIMSTAT?'` must answer `1` in the second field and `AT+CPIN?` must not return CME ERROR 10 or 13; otherwise reseat the card and `simei at 'AT+CFUN=1,1'`. On the `at` backend, another program may hold `/dev/ttyUSB3` or echo may be on; `simei esim info` sends `ATE0` first. On the `uqmi` backend, the feed's driver needs `LPAC_QMI_DEV`, which simei sets but the stock `/usr/bin/lpac` wrapper does not, and it opens its UIM client on a hardcoded `/dev/cdc-wdm0` (SFT-142). `at` stays the default until the doctor passes on the uqmi path.
- **Another eSIM operation is running**: only one lpac run is allowed at a time. If a run was killed, the next command clears the stale lock by itself; a genuinely parallel run (TUI and ssh at the same time) has to finish first.
- **Web TUI page empty in LuCI**: LuCI was opened over https; use `http://` or the direct link shown on the page.
- **Board unreachable after a DHCP change**: look up the new address (home router, or `tools/serial.sh run "ip -4 addr show br-lan"`) and update `tools/router.env`.